Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Learner Registration Number

Pearson BTEC Level 3 Nationals Certificate, Extended Certificate, Foundation Diploma, Diploma, Extended Diploma

Time 40 minutes

Paper reference **31617H/1B**

Applied Science/Forensic and Criminal Investigation

UNIT 1: Principles and Applications of Science I

Biology

SECTION A: STRUCTURES AND FUNCTIONS OF CELLS AND TISSUES

You must have:
A calculator and a ruler.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided – *there may be more space than you need.*

Information

- The exam comprises three papers worth 30 marks each:
  – Section A: Structures and Functions of Cells and Tissues (Biology)
  – Section B: Periodicity and Properties of Elements (Chemistry)
  – Section C: Waves in Communication (Physics).
- The total mark for this exam is 90.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

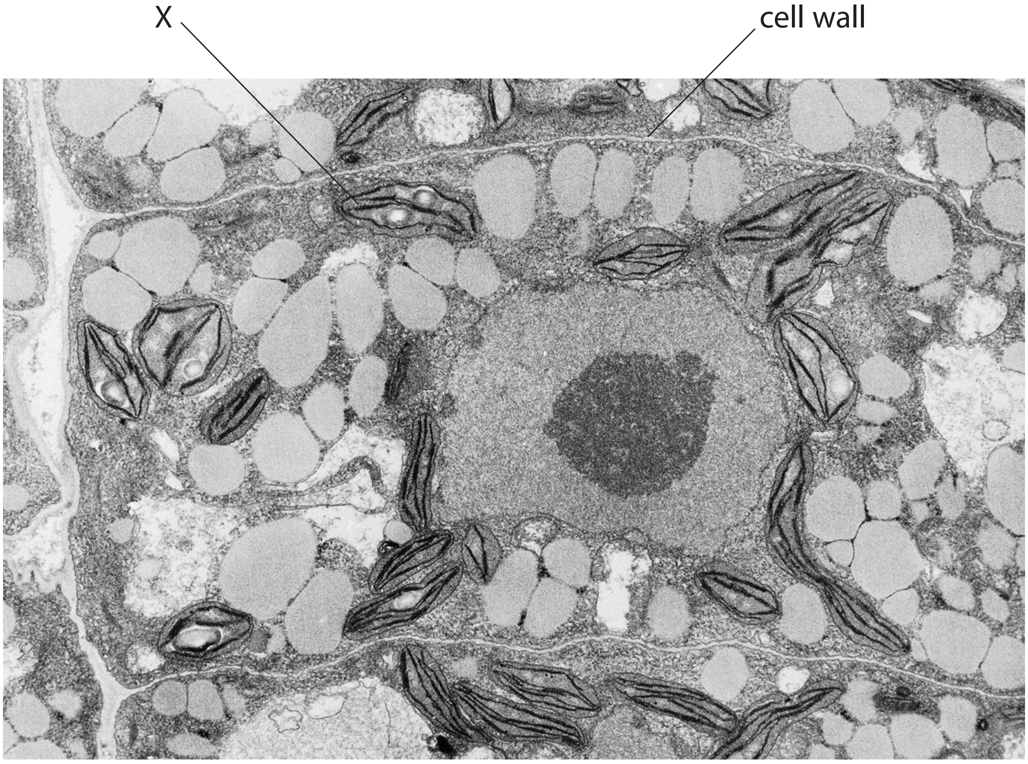
Turn over ▶

Pearson

Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

1 Figure 1 shows an electron micrograph of a plant cell and some surrounding cells.



(Source: ©PAL)

Figure 1

(a) Identify the cell component labelled X in Figure 1.

(1)

☐ **A** cell membrane

☐ **B** chloroplast

☐ **C** pit

☐ **D** tonoplast

(b) State the type of ribosome found in plant cells.

(1)

....

(c) Root hair cells are specialised plant cells.

Give **two** functions of root hair cells.

(2)

1 ....

2 ....

(d) A plant cell is viewed using a microscope.

The observed diameter of the plant cell is 13.2 mm.

The plant cell has an actual diameter of 66 μm.

Calculate the magnification used to view the image.

(3)

Show your working.

magnification = ... ×

(Total for Question 1 = 7 marks)

2 Figure 2 shows a diagram of a synapse in the brain.

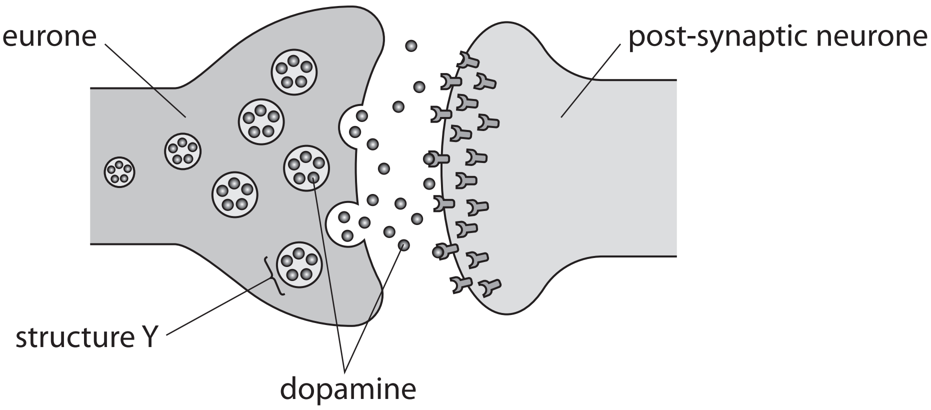


Figure 2

(a) Identify structure Y in Figure 2.

(1)

....

(b) Paragraph 1 partly describes the action of dopamine at the synapse.

Dopamine is a ....R.................. released from the pre-synaptic membrane via exocytosis. Dopamine crosses the cleft by the process of ....S.................. . Dopamine binds with ....T.................. on the post-synaptic membrane.

Paragraph 1

Identify the missing words R, S and T in Paragraph 1.

(3)

R..

S..

T..

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(c) An imbalance in dopamine levels can contribute to ill health.

(i) Name **one** disease that can be caused by an imbalance of dopamine.

(1)

....

(ii) Identify the precursor molecule which can be given to raise dopamine levels.

(1)

- ☒ **A** ATP
- ☒ **B** L-Dopa
- ☒ **C** muscarine
- ☒ **D** nicotine

(Total for Question 2 = 6 marks)

3 The lungs are part of the respiratory system.

(a) Gas exchange occurs in the alveoli in the lungs.

Describe the structure of the walls of the alveoli.

(2)

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DO NOT WRITE IN THIS AREA

(b) Figure 3 shows ciliated epithelial cells and goblet cells found in the lungs.

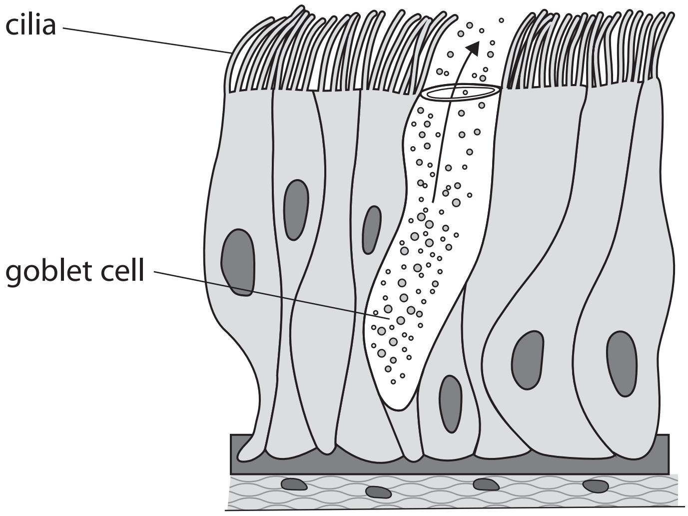

Figure 3

Explain how the ciliated epithelial cells and goblet cells protect the lungs.

(4)

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(Total for Question 3 = 6 marks)

4 Neutrophils and B-lymphocytes are white blood cells involved in the body's immune response to pathogens.

(a) (i) Figure 4 shows a diagram of blood cells at high magnification.

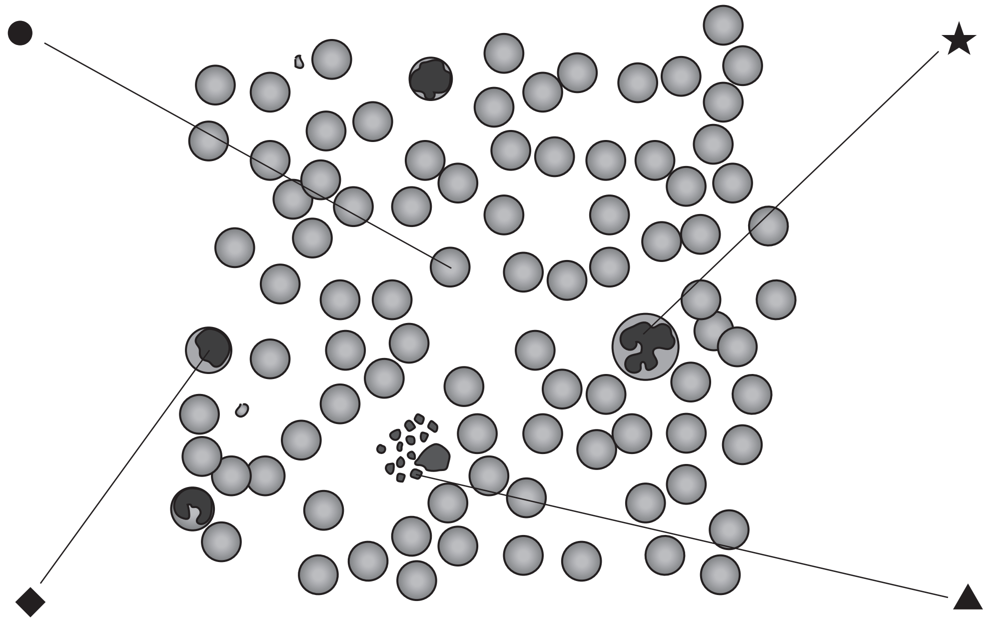

Diagram not to scale

Figure 4

Identify the symbol that shows a neutrophil in Figure 4.

(1)

☒ **A** ●

☒ **B** ◆

☒ **C** ▲

☒ **D** ★

(ii) Neutrophils bind to pathogens. Pathogens are ingested by phagocytosis.

Describe how pathogens are destroyed by neutrophils after they are ingested.

(2)

(iii) Explain the role of B-lymphocytes in the immune response.

(2)

(Total for Question 4 = 5 marks)

5 Figure 5a shows a diagram of a cell containing a nucleus.

Figure 5b shows a diagram of a cell containing a nucleoid.

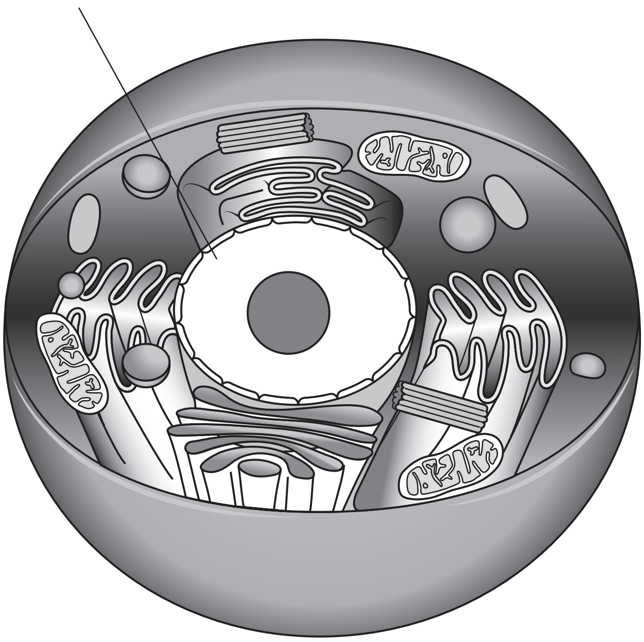

Figure 5a

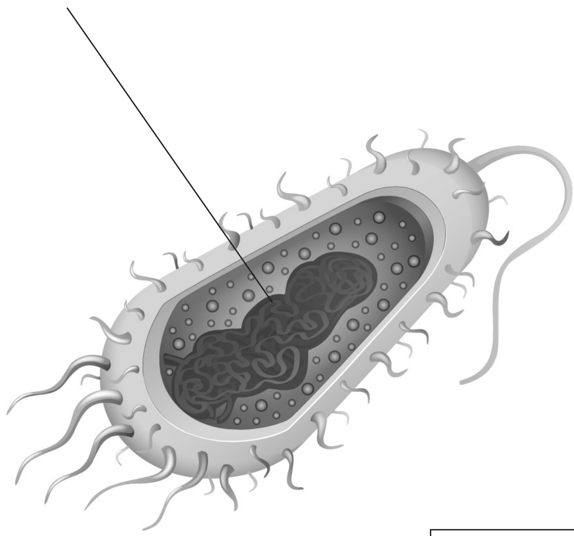

Diagram not to scale

(Source: ©PAL)

Figure 5b

Compare the structures and functions of a nucleus and a nucleoid.

Your answer should refer to similarities and differences.

(6)

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(Total for Question 5 = 6 marks)

TOTAL FOR PAPER = 30 MARKS

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